Continuation Plan

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Version Control

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0.3	3-12-2024	Added recommendations for the next project team
0.4	18-12-2024	Added credentials
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1. Introduction

This Continuation Plan serves as a comprehensive guide for the next team assigned to the Keuzewijzer project. The Keuzewijzer is an innovative tool designed to assist students in selecting their graduation pathways by providing clear insights into available profiles and course clusters. This document consolidates all essential information about the project's current state, tools, and processes to ensure a seamless handover.

The plan outlines the project goals, achievements, and areas for future development while addressing technical details, stakeholder expectations, and communication guidelines. By following this guide, the next team can build upon the foundation established, improve the tool's functionality, and enhance its value for students, advisors, and other users.

1. Project and purpose

The Keuzewijzer is an interactive tool designed to assist students in selecting a graduation pathway that aligns with their interests, skills, and career goals. By providing clear insights into the available profiles and the associated coursework or clusters required to complete them, the tool simplifies the decision-making process. It helps students understand the expectations and prerequisites of each option, enabling them to make informed choices that suit their personal aspirations and academic strengths.

1.1. Continuation plan

The continuation plan is set up for the next group tasked with working on the Keuzewijzer. While the basic functionality is already in place, there are many additional features that can be implemented to enhance its usefulness. These features include improving the Keuzewijzer and adding features that can benefit students, student advisors, and even teachers. In this document, we will provide advice on how to move to the next part of the project and more details will be described.

1.2. Goals

The primary goal of the Keuzewijzer tool is to facilitate a seamless decision-making process for students as they choose their graduation pathways. This involves providing comprehensive and tailored information that matches students' individual interests, skills, and career objectives. By doing so, the tool aims to ensure that each student can make an informed and confident decision about their academic future.

Looking forward, a strategic goal for the Keuzewijzer is to incorporate predictive analytics capabilities. This enhancement would allow the tool to forecast students’ likely choices of coursework clusters based on their historical preferences and academic performance trends. By predicting these selections, the tool can significantly assist schedule makers. This foresight enables schedule planners to proactively organize and optimize resources, thereby ensuring that necessary courses and materials are available ahead of actual student selections. This predictive capability not only streamlines the scheduling process but also enhances the overall educational planning experience for both students and administrators.

2. Current State Assessment

The project has come quite a long way, but unfortunately, we weren't able to completely finish the project. In this segment we will talk about what we were and weren't able to realize for the project.

What we did manage to complete is the different majors that students are able to follow on Windesheim Zwolle. Furthermore, did we manage to create a working dashboard in which the maintainers are able to create new majors, cohorts, clusters and accounts. The maintainers of this dashboard also have the possibility to completely edit each cluster and major.

2.1. Stakeholders

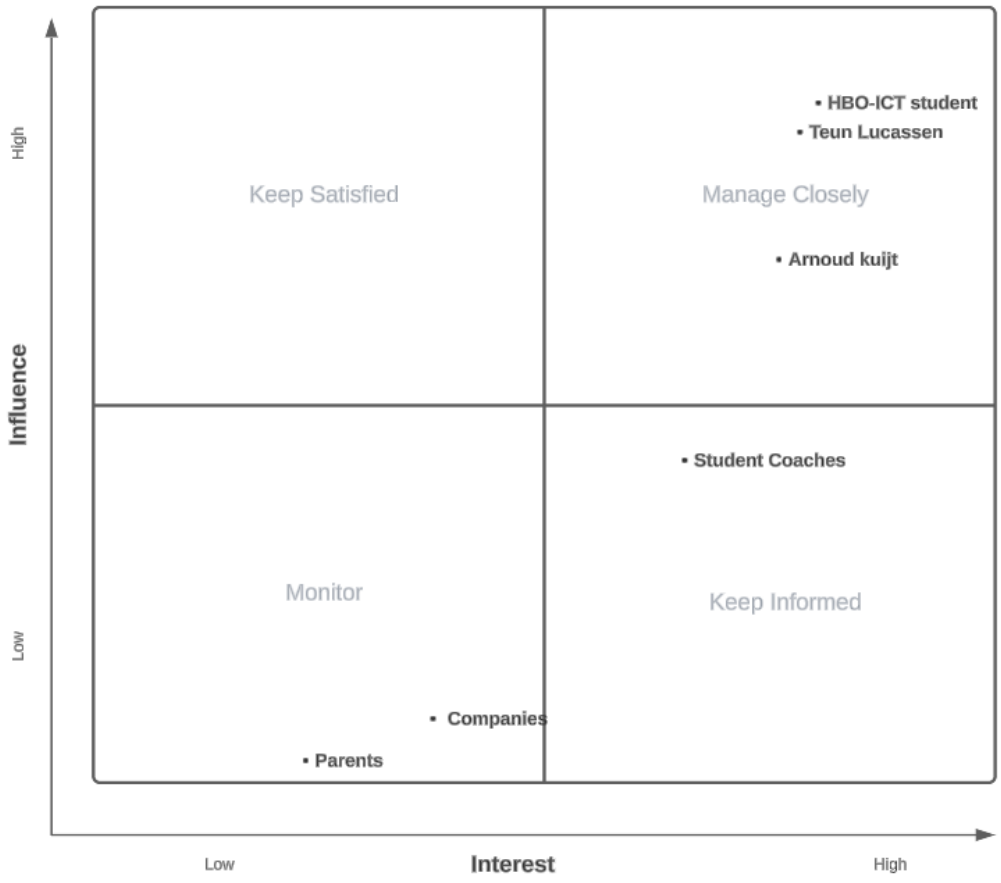


Figure 1: Stakeholder analysis

Stakeholder	Involvement
Teun Lucassen	Product Owner of the project.
Arnoud Kuijt	Learning route advisor

Student Coaches	Student coaches will be using the tool to help students choose a path for the certificate they are going to receive when finishing their degree. That is also why student coaches will be one of the most frequent users of this tool.
HBO-ICT student	HBO-ICT students will be using it to see what path they can follow for the degree they want to get. The students can use this tool whenever they want to look up and see if a course fits their expectations and will be the most frequent users of this tool.
Companies	Windesheim shows the Keuzewijzer to companies so that companies are able to discern what level the students have acquired.
Parents	Parents can monitor their children's learning progress on Windesheim using this tool.

2.2. Client contact

Name	Contact
Teun Lucassen	t.lucassen@windesheim.nl
Arnoud Kuijt	a.kuijt@windesheim.nl

2.3. Resources

During the project the use of tools and technologies helped to build, manage and deploy the Keuzewijzer.

2.3.1. Tools

2.3.1.1. Gitlab

The Keuzewijzer utilizes Gitlab to manage the project's source code. Gitlab facilitates tracking changes in branches and validating risks. Compared to GitHub, Gitlab was selected because Windesheim has its own instance of Gitlab. This allows the project repository to be managed by individuals, such as students, who can work on the project using their student accounts.

2.3.1.2. IDEA's

The IDEA's that are used depend on the person working on the project. The group that started the new Keuzewijzer worked in Visual Studio Code and Webstorm from Jetbrains. These IDEA's help build and test the Keuzewijzer.

2.3.2. Technologies

2.3.2.1. NextJS

For the frontend of the Keuzewijzer project, including both the main application and the dashboard, Next.js is used to build the interface. Next.js is a React-based JavaScript framework known for its powerful features like server-side rendering (SSR) and static

site generation (SSG), which help improve performance and SEO. This is especially useful for Keuzewijzer, where fast loading times and good search engine visibility are important. Next.js also simplifies routing with its file-based system, making it easy to manage and scale the application as it grows. Overall, Next.js provides excellent performance, flexibility, and a streamlined development experience, making it a great fit for building the dynamic and user-friendly interface of Keuzewijzer.

2.3.2.2. ASP.NET

For the backend of the Keuzewijzer project, ASP.NET is used to create a REST API that handles server-side logic and data management. ASP.NET is a powerful web framework known for its robustness, scalability, and security features. By using ASP.NET's capabilities, the Keuzewijzer backend provides a robust API for the frontend to interact with, handling tasks such as data retrieval, user authentication, and management of user-specific information. The framework's built-in tools for handling database migrations, serialization, and authentication help streamline the development process. ASP.NET also offers strong security features, including protection against common vulnerabilities like SQL injection and cross-site scripting (XSS). With its ability to scale and its efficient handling of backend tasks, ASP.NET is an ideal choice for building the REST API that powers the Keuzewijzer application.

2.3.2.3. CI/CD Pipeline

For the Keuzewijzer project, the CI/CD pipeline in GitLab is configured to run only when code is pushed to the main branch. The pipeline checks for errors, builds the front-end and back-end components, and deploys them to the appropriate environments. This ensures that only stable, tested code is deployed to production, while allowing the team to focus on the development of features without triggering deployments for every change.

2.3.2.4. Google Analytics

To track and analyze user interactions the Keuzewijzer uses Google Analytics. Google Analytics collects data on user behavior, such as page views, session duration, and navigation patterns, Google Analytics helps identify areas for improvement in the user interface and experience. This data-driven approach enables the team to make informed decisions about how to enhance the application, ensuring it meets user needs and performs effectively.

3. Project Roadmap

3.1. Done functionality

The most desired functionalities have been developed and are in use within the Keuzewijzer. Refer to Functional Design to see all the functionalities that have been implemented.

3.2. Future tasks

The potential future tasks for the Keuzewijzer are outlined below. Unfortunately, we were unable to implement these functionalities due to time constraints. Nevertheless, please remember to consult the client regarding the desired functionalities for the Keuzewijzer, as there may be new changes or insights that the client prefers.

Retake option for clusters

- Make it so that students can select retake on clusters so that the path adapts to the delay in the study route.
- The retaken clusters get duplicated to the next possible location
- The path updates according to the defined rules which are: level, 1st semester/2nd semester, prerequisites.
- This is almost implemented. When a cluster is being displayed (either if it's the only option available or is the selected option from several to take) the cluster attribute "chosen" must be set as "True". If a cluster wants to be retaken, the attribute "retake" must be added one. Then the route should be sent to the backend and processed first by the function "decompress_validate" and then by "transform_clusters" and this way it will be rebuilt considering that retake.

Free choice semester

- Add a list of clusters that can be chosen for the free choice semester
- There is an available endpoint that already returns all the possible clusters to retake. This must be filtered and processed

Keuzewijzer for Parttime Students at Windesheim Zwolle

- Adapt the existing Keuzewijzer to meet the needs of part-time students at Windesheim Zwolle.

- Include flexible scheduling options, accounting for students who balance studies with work.
- Ensure support for modular study paths and long-term planning.
- Provide specific guidance tailored to the part-time program's curriculum and clusters.

Keuzewijzer for Fulltime Students at Windesheim Almere

- Customize the Keuzewijzer for the full-time students at Windesheim Almere.
- Incorporate the Almere campus curriculum and program offerings.
- Make it understandable for the students at Almere what choices they have in Zwolle, what prerequisites there are and what they can do in Almere.

Prediction System for the popularity of clusters

- Develop an AI-based prediction system to forecast which clusters will be the most selected by students.
- Make forecasts based on the tracked activity of the students on the web, the different selected routes or combining individual predictions for each student to create a combined estimate.
- Help school with class and room scheduling by providing data that helps with predicting how many students will choose what clusters.
- Continuously improve prediction accuracy by training with the activity and results of the previous year

AI application for helping students

- Implement an AI-powered application to suggest to students what are the standard or most popular routes or making suggestions on the route choices based on the general activity.
- Provide context of the current clusters. Feeding a RAG model the documentation about each cluster, a chatbot could be built that explains and provides context to the students about the different clusters.
- Example use cases: "What cluster should I follow if I want to specialize in AI?" or "What courses are part of the Data Science cluster?"
- Also provide most popular/suggested free choices for your chosen path.

Make Graduation Possible Without a Profile

- Redesign the graduation requirements to offer students the flexibility to graduate without adhering strictly to a predefined profile.
- Allow for a more customized and interdisciplinary study path.

- Ensure that the Keuzewijzer reflects these changes and supports students in planning a profile-free graduation trajectory.

Downloading Your Study Route

- Enable students to download their customized study route as a PDF or other formats.
- Include key details such as completed courses, planned courses, clusters, and milestones.
- Ensure the downloaded route is visually appealing and easy to understand.
- Add options for customizing the export (e.g., choosing to include only specific sections).

Sharing Your Study Route

- Implement functionality for students to share their study routes with others, such as peers, mentors, or academic advisors.
- Add options to share via email, social media, or direct link.
- Include privacy controls to allow students to choose what information they want to share.
- Facilitate collaborative planning, enabling advisors to provide feedback on shared study routes.

3.3. Known issues

On the issues board of GitLab, there is 1 unresolved bug. When you pick a graduation semester and you make the screen smaller, then you have to pick again.

3.4. Technical debt

With the intention of faster developing, the ordering algorithms have not been tested in a standardized way. The file tests.py inside the backend represents a replication of the retake and ordering logic that exists inside the ClusterService but the optimal way to perform this would be to create proper unit tests for each function that cover a more extensive number of situations.

The idea was also to do cypress tests which haven't been done so these still need to be done to make sure the frontend has been tested completely. This can be done together with the lighthouse tests which have been done.

3.5. Learned lessons

Faced challenges

One of the biggest challenges in this project was to create a logic system that works under the current rule system but is dynamic enough to be easily adapted in case new clusters or new set of rules are created in the upcoming years. The implemented solution fits both these criteria and also integrates in a simple and accessible way with the frontend, so that both can work in an independent way and may be exchangeable between different versions of the updated code.

Potential risks

Even if the current version tries to take into account possible rule addition and new clusters in the ordering and display of them, if there is a drastic change of the rules it will be necessary to change the whole clusterService in charge of the logic. Apart from the change of the business logic the application has been built with standardized packages and stable versions so there should not be any risk of depreciation in the short to medium future. The web must be updated according to the internet standards in order to maintain its safety guarantees.

Best practices

In order to keep the product as simple and accessible as possible the different functionality both for the backend and for the frontend must be included inside different services that make it organized and accessible for future developers. Also, the existing functionality should be used whenever it is applicable in order to have a centralized implementation of the functionality.

GitHub branches must be respected when developing new functionality for the product in order to safely maintain the live product away from the different development environments

4. Documentation

4.1. Analyze documents

We have made a very extensive package selection about which frameworks we are going to use for the solution. Within this package selection process, we set up requirements for front-end and back-end frameworks. These requirements were then used to score each of the researched frameworks to make a shortlist, then the frameworks in the shortlist were researched on a deeper level so that it was possible to make a substantiated choice and give the Product Owner a recommendation(s).

4.2. Design documents

All the functionality that has been realized can be found in Functional Design. This document will help understand all the actors, requirements, use cases and designs of

the Keuzewijzer. All the designs are made in Figma, based on the requirements, feedback from testing with students and wishes of the client.

For more complex and technical aspects of the project, refer to the Technical Design document. This document outlines the technical structure of the Keuzewijzer, such as what frameworks are used, packages and how the client and server are set up.

4.3. Manage documents

To keep the GitLab repository nice, clean and organized we made a document in which we explain a number of rules. These rules consist of naming conventions for branches, which branches are protected and those who are not, and a template to create an informational merge request description.

4.4. Tests

All the functionality testing has been described in the “test report” document. This can be found in the folder ‘Realize’.

4.5. Setup project

For the setup of the project, you need to get access to the GitLab repository of the Keuzewijzer. Ask the client for access to the repository. In this repository you can find the “read.me” file. In this file everything is described on how to start the project running in your local environment. After you make new functionality, reach out to the GitLab rules document to see if you have worked according to this document.

5. Communication Guidelines

These are some ways we did communication which you can look at and use as inspiration to decide how you will do it.

5.1. Preferred Communication Channels

1. **Team Communication**:
    a. Primary: WhatsApp, Discord, Teams or Slack for quick discussions and collaboration. We used WhatsApp and Teams.
    b. Sprint Retrospective: Use Metro Retro for interactive feedback and improvement sessions.
2. **Communication with the Product Owner**:
    a. Weekly meetings for progress updates and feedback.
    b. Secondary: Email for sharing reports or important documents.
    c. Always available for questions through teams or email.
3. **Task Management**:
    a. Use the GitLab Issue Board for managing user stories and tracking tasks.

5.2. Documentation Location

Decide on a central repository for documentation:

Options: Microsoft Teams, GitLab Wiki, Confluence, or Jira.

Meeting Cadence

Daily Standups:
Short internal team meetings to discuss progress, blockers, and next steps.
Weekly Product Owner Meetings:
Update the product owner on the sprint progress and clarify any priorities or questions.
Sprint Retrospective:
Conducted at the end of each sprint using Metro Retro to gather team feedback and discuss improvements.

6. Onboarding Checklist

- **Getting Started Tasks**: Create a checklist for new team members to follow when they join, such as setting up accounts, accessing systems, and reviewing documentation.

ITEM	Check
Request access to the GitLab repository for the Keuzewijzer project.	
Ensure the new team members have accounts on GitLab, Teams, or other relevant tools.	
Obtain credentials for Strato server (SSH access for managing Docker images).	
Obtain credentials for Google Analytics account for tracking user interactions.	
Obtain credentials for	
Clone the project repository from GitLab using the provided URL.	
Follow the instructions in the README.md file for setting up the project locally	
Install necessary dependencies	
Run backend and frontend components locally	
Verify that the CI/CD pipeline functions correctly after setup.	
Familiarize yourself with Functional Design document (for implemented features and user stories).	
Familiarize yourself with Technical Design document (for architectural details and frameworks used).	
Familiarize yourself with Test Report (to understand the testing procedures and coverage).	
Familiarize yourself with GitLab rules document (branching conventions, merge request templates, etc.).	

7. Next Steps

Resolve Known Issues

- Check the GitLab Issues Board for unresolved bugs or pending tasks.
- Prioritize fixing critical bugs or addressing technical debt that could hinder future development.

Plan for New Features

- Collaborate with the product owner to confirm priorities for upcoming features, such as:
    - Supporting part-time students at Windesheim Zwolle:
    - Expanding functionality for full-time students at Windesheim Almere.
    - Adding predictive analytics to predict the number of students that will take on a course by:
      - Registering the click activity throughout the site to see the most popular clusters in an approximate way.
      - If users log in with credentials, we can predict their most likely route based on their activity through the site or on their last selected route.
    - Suggesting recommended options for the hesitant students :
      - Storing the different built routes by everyone to process them as unlabelled data and extract underlying patterns. Training a model to suggest standard or popular learning routes or to suggesting alternatives to complete any study route.

8. Additional information

8.1. Credentials

8.1.1. Strato Server

To access the server of the Keuzewijzer, on which the Docker images of the project are running you need access to it with SSH. With SSH you can connect to the server and see all the files, docker images and more. For example, the old Keuzewijzer is also running on the Strato server, but not in a Docker image. For the credentials of the Strato server, ask the client.

8.1.2. Connect Gitlab to GitHub Desktop

If you want to work with Gitlab Desktop to work with branches and commit with a better user experience, you can make an access token on Gitlab.

1. Go to gitlab.windesheim.nl
2. Go to the Keuzewijzer project
3. Go to ***settings > Access Tokens***
4. Click on 'Add new token'
5. Fill in the information
    a. Token name (Your name for example)
    b. Expiration date (Set as long as needed for the project)
    c. Select a role (Select the role you want to give yourself)
    d. Select scopes (Check all)
6. Click on 'create token'.
7. Save the token
8. Go to GitHub Desktop
9. Clone the repository with the URL of the Keuzewijzer (https://gitlab.windesheim.nl/org.hbo-ict/keuzewijzer/wa_24-25_sem1/keuzewijzer.git)
10. Fill in your username of Gitlab and then the password is the token you created.
11. Now you have access to Gitlab using GitHub Desktop

8.1.3. Access to the dashboard

For the Keuzewijzer we've built a dashboard to manage the Keuzewijzer's clusters, graduate profiles, cohorts and more. For that you need access. Ask the client to make an account for you to get access to the dashboard.

8.1.4. Google Analytics

For the Keuzewijzer we've used Google Analytics to keep track of user's behavior on the website. Test the SEO and other things related to make the user experience of the

website better. You can get access to Google Analytics by asking the client if he can invite you to the property.